

MA2108

AY25/26 Sem 1

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Ordered Fields

D1.1 Field A field is a set F with addition and multiplication. $\mathbb{R}, \mathbb{Q}$ are fields while $\mathbb{N}, \mathbb{Z}$ are not.

Addition Axioms Closure of Addition, Commutativity, Associativity, Identity Element, Additive Inverse

Multiplication Axioms Closure of Multiplication, Commutativity, Associativity, Identity Element, Multiplicative Inverse, Distributive Law

P1.4

- $x + y = x + z \Rightarrow y = z$
- $x + y = x \Rightarrow y = 0$
- $x + y = 0 \Rightarrow y = -x$
- $-(-x) = x$
- $x \neq 0 \wedge xy = xz \Rightarrow y = z$
- $x \neq 0 \wedge xy = x \Rightarrow y = 1$
- $x \neq 0 \wedge xy = 1 \Rightarrow y = 1/x$
- $x \neq 0 \Rightarrow 1/(1/x) = x$
- $0x = 0$
- $x \neq 0 \wedge y \neq 0 \Rightarrow xy \neq 0$
- $(-x)y = -(xy) = x(-y)$
- $(-x)(-y) = xy$

D1.5 Total Order A relation satisfying trichotomy and transitivity property. An ordered set is a set with a total order.

D1.8 Ordered Field A field where $y < z \Rightarrow x + y < x + z$,

$x > 0, y > 0 \Rightarrow xy > 0$. All ordered fields contain (a homomorphism of) $\mathbb{Q}$.

P1.12 Ordered Field Statements

- $x > 0 \Rightarrow -x < 0$ and vice versa
- $x > 0 \wedge y < z \Rightarrow xy < xz$
- $x < 0 \wedge y < z \Rightarrow xy > xz$
- $xy > 0 \Rightarrow (x > 0 \wedge y > 0) \vee (x < 0 \wedge y < 0)$
- $x \neq 0 \Rightarrow x^2 > 0$
- $0 < x < y \Rightarrow 0 < 1/y < 1/x$

P1.15 There exists $\iota : \mathbb{Q} \hookrightarrow F$ such that ι is order preserving for all ordered fields.

Least Upper Bound Property

D1.16 Upper Bound Suppose S is an ordered set and $E \subset S$. If

$\exists \beta \in S, \forall x \in E, x \leq \beta$, then E is bounded above and β is an upper bound of E . Lower bound is defined similarly with $\geq$. They might not be unique and could possibly not exist. They might not be in the set itself.

D1.19 Least Upper Bound Suppose $\alpha \in S$, α is an upper bound of E , and if $\gamma < \alpha$ then γ is not an upper bound of E . Then α is the least upper bound or supremum of E , and $\alpha = \sup E$. We can similarly define the greatest lower bound or infimum $\alpha = \inf E$. The LUB is also unique.

R1.20 Maximum α is a maximum of E if it is a LUB and $\alpha \in E$.

T1.26 Least Upper Bound Property An ordered set S where $\forall E \subset S, E \neq \emptyset$, and E is bounded above implies $\exists \sup E \in S$. $\mathbb{Q}$ does not have this property.

T1.29 Completeness of $\mathbb{R}$ $\mathbb{R}$ is the ordered field which has the LUB property and is unique up to isomorphism. It also contains $\mathbb{Q}$ as a subfield.

T1.32 An ordered set with the LUB property also has the GLB property.

P1.34 Let $E \subset \mathbb{R}, a \in \mathbb{R}, T = \{a + x : x \in E\}$. If E is bounded above, then $\sup T = a + \sup E$.

• $\emptyset \neq E \subset \mathbb{R}, E$ is bounded above, so $\sup E$ exists

• $\forall x \in E, x \leq \sup E, a + x \leq a + \sup E$

• $a + \sup E$ must be an upper bound for T , let M be any upper bound of T

• $\forall x \in E, a + x \leq M, x \leq M - a$

• $M - a$ is an upper bound for E

• By definition of supremum,

$$\sup E \leq M - a, a + \sup E \leq M, \sup T \geq a + \sup E$$

R1.35 To show equality, either show $A \leq B \wedge B \leq A$ or $A \leq B, A < B$ is a contradiction

Archimedean Property

D2.1 Archimedean Property Let F be an ordered field. F has the Archimedean Property if $\forall a, b \in F, a > 0, \exists n \in \mathbb{N}, na > b \in F$

D2.1 Density For an ordered field $F, \mathbb{Q}$ is dense in F wrt ordering if

$$\forall x, y \in F, x < y, \exists r \in \mathbb{Q}, x < r < y \in F$$

P2.4 F has the Archimedean Property iff $\mathbb{Q}$ is dense in F

$\mathbb{Q}, \mathbb{R}$: $\mathbb{Q}$ is Archimedean and is dense in itself, while $\mathbb{R}$ is Archimedean as well

C2.6 $\forall \varepsilon \in \mathbb{R}_{>0}, \exists n \in \mathbb{N}, 0 < \frac{1}{n} < \varepsilon$

T2.7 nth roots $\forall x \in \mathbb{R}_{>0}, n > 0 \in \mathbb{Z}, \exists ! y, y^n = x$

C2.8 For positive real a, b and positive integer $n, (ab)^{1/n} = a^{1/n}b^{1/n}$

C2.9 If $a, b \in \mathbb{R}, a < b, \exists x \in \mathbb{R} \setminus \mathbb{Q}, a < x < b$

P2.10 Let $\alpha \in \mathbb{R}, E = \{x \in \mathbb{Q} : x < \alpha\} \subseteq \mathbb{Q}$. Then $\sup E = \alpha$

Sequences

P2.13 Properties of absolute value

- $|a| \geq 0, a \leq |a|, -a \leq |a|$
- $|a| = 0 \Leftrightarrow a = 0$
- $|-a| = |a|$
- $|ab| = |a| \cdot |b|$
- $|a|^2 = a^2$
- $c \geq 0 \Rightarrow |a| \leq c \Leftrightarrow -c \leq a \leq c$
- $-|a| \leq a \leq |a|$

P2.14 Triangle Inequality $\forall a, b \in F, |a| + |b| \leq |a| + |b|$

C2.15 $\forall a, b \in F$,

- $||a| - |b|| \leq |a - b|$
- $|a - b| \leq |a| + |b|$
- $||a| - |b|| \leq |a + b|$

D2.16 Sequences A sequence in a set X is a function $x : \mathbb{N} \rightarrow X$, denoted by $(x_n)_n$ or $\{x_n\}_n$.

D2.21 Convergence A sequence in F converges iff

$\exists x \in F, \forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow |x_n - x| < \varepsilon$. For every small distance ε , there is a cut-off N such that for all larger n , the absolute difference is smaller than ε . x is the limit of the sequence in F , and $\lim_{n \rightarrow \infty} x_n = x$.

P2.23 The limit of a sequence is unique.

D2.24 Bounds A sequence is eventually constant if

$\exists N \in \mathbb{N}, \forall n \geq N, x_n = x_N = x$. A sequence is bounded in F if the set of values is bounded in F by some B . We can define one sided bounds similarly.

P2.26 If a sequence is convergent in F , then it is bounded.

E2.29 Example Convergence Proof (Make n subject of formula)

$$\begin{aligned} \left| \frac{2n^2 + 1}{n^2 + 3n} - 2 \right| &= \left| \frac{2n^2 + 1 - 2(n^2 + 3n)}{n^2 + 3n} \right| \\ &= \left| \frac{1 - 6n}{n^2 + 3n} \right| \\ &\leq \frac{1 + 6n}{n^2 + 3n} \\ &\leq \frac{n + 6n}{n^2} = \frac{7n}{n^2} = \frac{7}{n} \end{aligned}$$

• Let $\varepsilon > 0$ be given, then choose $K \in \mathbb{N}$ such that $K > \frac{7}{\varepsilon}$. Then $\forall n \geq K$,

$$\begin{aligned} \left| \frac{2n^2 + 1}{n^2 + 3n} - 2 \right| &\leq \frac{7}{n} \text{ since } n \geq K \\ &\leq \frac{7}{K} \\ &< \varepsilon \text{ since } K > \frac{7}{\varepsilon} \end{aligned}$$

Convergent Sequences

P3.1 Properties of Limits of Convergent Sequences Let $(s_n)_n, (t_n)_n$ be convergent sequences in F . Then

$(s_n + t_n)_n, (cs_n)_n, (s_nt_n)_n, (|s_n|)_n, (\sqrt{s_n})_n, (\frac{1}{s_n})_n$ are all convergent with limits following the substitution. If $\exists N \in \mathbb{N}, \forall n \geq N, s_n \leq t_n$, then $\lim_{n \rightarrow \infty} s_n \leq \lim_{n \rightarrow \infty} t_n$.

P3.2 Squeeze Theorem If $(x_n)_n, (y_n)_n, (z_n)_n$ are sequences in an ordered field F such that $\forall n, x_n \leq y_n \leq z_n$ and $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} z_n = a$, then $\lim_{n \rightarrow \infty} y_n = a$.

More Sequences

E3.3 For real $0 \leq b < 1, \lim_{n \rightarrow \infty} b^n = 0$.

E3.6 For real $c > 0, \lim_{n \rightarrow \infty} c^{\frac{1}{n}} = 1$.

E3.7 For real $a > 1$ and integer $k > 0, \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0$.

E3.8 $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$

D3.9 Monotone Sequences (s_n) is monotonically increasing if $\forall n, s_n \leq s_{n+1}$, while it is monotonically decreasing if $\forall n, s_n \geq s_{n+1}$.

T3.10 Monotone Convergence Every monotone sequence in $\mathbb{R}$ is bounded iff it converges.

Subsequences, Cauchy Sequences

D4.2 Subsequence Given a sequence in F , consider $(n_k)_k$ of positive integers where $n_1 < n_2 < \dots < n_k < \dots$. Then $(x_{n_k})_k$ is a subsequence of $(x_n)_n$.

E4.3 A sequence converges to x iff every subsequence also converges to x .

R4.5 For divergence, we can show a subsequence which diverges, or two subsequences which converge to different limits.

T4.9 Existence of Monotone Subsequence Let there be a sequence in an ordered field F which is not necessarily convergent. There exists a monotone subsequence in it.

T4.10 Bolzano-Weierstrass Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

D4.11 Cauchy Sequence A sequence $(p_n)_n$ in F is said to be Cauchy if

$$\forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall m, n \in \mathbb{N}_{\geq N}, |p_n - p_m| < \varepsilon.$$

P4.13 If a sequence is convergent, then it is Cauchy.

P4.15 If a sequence is Cauchy, then it is bounded.

P4.16 Properties of Cauchy Sequences If $(s_n), (t_n)$ are Cauchy sequences, then $(s_n + t_n), (cs_n), (s_nt_n), (|s_n|)$ are also Cauchy.

More Cauchy Sequences

D4.18 Cauchy Complete An ordered field in F is Cauchy complete iff every Cauchy sequence in F is convergent in F . $\mathbb{R}$ is Cauchy complete.

P4.19 Let p_n be a Cauchy sequence in F . If there exists a subsequence of p_n that converges in F , then p_n also converges in F to the same limit.

T4.20 $\mathbb{R}$ is Cauchy complete.

D4.22 Contractive A sequence x_n is contractive if there exists C with $0 < C < 1$ such that $|x_{n+2} - x_{n+1}| \leq C|x_{n+1} - x_n|$ where C does not depend on n .

P4.23 Every contractive sequence is Cauchy and thus convergent.

D4.24 Extended Real Numbers The extended real number system consists of $\mathbb{R}, +\infty$, and $-\infty$. The original order is preserved and $\forall x \in \mathbb{R}, -\infty < x < +\infty$.

This ordered set (not field) is denoted by $[-\infty, +\infty]$. The supremum and infimum of any subset always exists.

D4.28 Limit to Infinity The limit of x_n is $+\infty$ if for all real $M, \exists N, \forall n \geq N, s_n \geq M$. Similarly for $-\infty$.

P4.30 Let x_n be a sequence in $\mathbb{R}, x \in [-\infty, +\infty]$. Then $x_n \rightarrow x$ iff $\forall A \in [-\infty, \infty], A < x, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow A < x_n$ and the opposite holds.

L4.31 For any sequence $x_n \in \mathbb{R}$, there exists a subsequence x_{n_i} which converges in $[-\infty, +\infty]$.

limsup, liminf

D5.1 limsup, liminf Let x_n be a sequence in $\mathbb{R}$. Let $E = \{\text{the limits in } [-\infty, +\infty] \text{ of all convergent subsequences of } x_n\}$. Then define $\limsup_{n \rightarrow \infty} = \sup(E)$, $\liminf_{n \rightarrow \infty} = \inf(E)$. $\limsup$ is the limit superior, while $\liminf$ is the limit inferior.

P5.3 Equivalent Definition Let x_n be a bounded series in $\mathbb{R}$ and $M = \limsup x_n$. $M = \limsup x_n$ if $M \in \mathbb{R}$ satisfies the following: For each $\varepsilon > 0$ there are at most finitely many n such that $x_n \geq M + \varepsilon$. Alternatively, $\forall \varepsilon, \exists K \in \mathbb{N}, \forall n \geq K, x_n < M + \varepsilon$. Also, for each $\varepsilon > 0$, there are infinitely many n such that $x_n > M - \varepsilon$. The same holds for $\liminf$.

P5.5 Convergence with limsup, liminf Let s_n be a sequence in $\mathbb{R}$. Then $\liminf_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} s_n \in [-\infty, +\infty]$. Equality only holds iff s_n converges.

P5.7 If $s_n \leq t_n, \forall n \geq N, \liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n$ and $\limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n$

P5.8 Equivalent Definition of limsup Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $y_n = \sup\{x_k : k \geq n\}$. Then $\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$.

P5.9 Equivalent Definition of liminf Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $z_n = \inf\{x_k : k \geq n\}$. Then z_n is increasing and bounded and $\liminf x_n = \lim_{n \rightarrow \infty} z_n$

Series

D5.10 Series Let F be an ordered field. Given a sequence $a_n \in F$, we can form an infinite series $\sum_{k=1}^{\infty} a_k = a_1 + a_2 + \dots$. More precisely, define $S_k = \sum_{n=1}^k a_n$ (nth partial sum).

P5.13 Properties of Convergent Series If $\sum a_n, \sum b_n$ are convergent series and $c \in F$, then $\sum (a_n + b_n)$, and $\sum ca_n$ are convergent.

P5.14 Cauchy Criterion for Series Let a_n be a sequence in $\mathbb{R}$. Then $\sum a_n$ converges iff $\forall \varepsilon > 0, \exists N, \forall m, n, m \geq n \geq N, |\sum_{k=n}^m a_k| \leq \varepsilon$.

C5.15 If $\sum a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$. However, the converse is false.

C5.17 nth term test If a_n does not converge or converges to a nonzero, then $\sum a_n$ diverges.

D5.19 A series eventually has a property (*) if $\exists N \in \mathbb{N}, \forall k \geq N, a_k$ has (*).

P5.20 Let $\sum a_n$ be an eventually non-negative series where $a_n \in \mathbb{R}$. Then a_n converges iff S_n is bounded above.

E5.21 Geometric Series For $\sum_{k=1}^{\infty} ar^{k-1}, s_n = \frac{a(1-r^n)}{1-r}$ if $r \neq 1$.

E5.22 p-series If $p > 1$, the p-series $\sum \frac{1}{n^p}$ converges. When $p \leq 1$, the series diverges.

D5.23 Absolute Convergence The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. If a series converges but not absolutely, it converges conditionally.

P5.24 Let a_n be a sequence in $\mathbb{R}$. If $\sum a_n$ converges absolutely, then a_n converges. However, the converse is false.

Convergence Tests

P6.1 Direct Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_k, \sum b_k$ are eventually non-negative. Suppose we eventually have that $0 \leq a_k \leq b_k$. If $\sum b_k$ converges, then $\sum a_k$ converges. If $\sum a_k$ diverges, then $\sum b_k$ diverges.

P6.4 Limit Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_n, \sum b_n$ are eventually positive, and suppose the limit $\rho = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$

exists. If $\rho > 0$, they either both converge or both diverge. If $\rho = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

P6.5 Ratio Test Let a_n be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually positive, and let $\rho = \limsup \frac{a_{n+1}}{a_n}$. If $\rho < 1$, $\sum a_n$ converges. If $\exists N \in \mathbb{N}, \forall n \geq N, \frac{a_{n+1}}{a_n} \geq 1$, then the series diverges.

P6.8 Root Test Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually non-negative, and let $\rho = \limsup a_n^{1/n}$. If $\rho < 1$, then $\sum a_n$ converges. If $\rho > 1$, then $\sum a_n$ diverges. If $\rho = 1$, we cannot make a conclusion.

L6.9 For any sequence c_n of positive numbers, $\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}$.

T6.11 Dirichlet's Test Let a_n, b_n be sequences in $\mathbb{R}$. Suppose the partial sums A_n form a bounded sequence, b_n is monotone decreasing, and $\lim b_n = 0$. Then $\sum a_n b_n$ converges.

L6.12 Partial Summation Given a_n, b_n , indexed by $Z \geq 0$, define $A_{-1} = 0$ and $\forall n \geq 0, A_n = \sum_{k=0}^n a_k$. Then for any $0 \leq p \leq q$, we have $\sum_{q=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q B_q - A_{p-1} b_p$.

E6.13 $\frac{\cos(n)}{\sqrt{n}}$ is convergent using Dirichlet's and $\sum_{k=1}^n \cos k = \frac{\sin(n+\frac{1}{2})-\sin \frac{1}{2}}{2 \sin \frac{1}{2}}. \sum_{k=1}^n \sin k = \frac{\sin(\frac{n}{2}) \sin(\frac{n+1}{2})}{\sin \frac{1}{2}}.$

C6.14 Alternating Series Test Let $\sum (-1)^{n+1} b_n$ or $\sum (-1)^n b_n$ be an alternating series with b_k as a sequence in $\mathbb{R}$. Suppose $b_n \geq 0, b_n$ is monotone decreasing, and $\lim_{n \rightarrow \infty} b_n = 0$. Then it is convergent.

Rearrangement of Series

D6.15 Rearrangement Let σ be a permutation of $\mathbb{N}$. Then the series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$.

T6.16 Rearrangement of Absolutely Convergent Series Let a_n be a sequence in $\mathbb{R}$, and $\sum a_k$ be an absolutely convergent series. Then any rearrangement of $\sum a_k$ also converges and has the same sum as $\sum a_k$.

T6.17 Riemann Rearrangement Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is conditionally convergent. Let $\alpha, \beta \in [-\infty, +\infty], \alpha \leq \beta$. Then there exists a rearrangement $\sum b_n$ of $\sum a_n$ such that the sequence of partial sums s'_n of b_n satisfies $\liminf s'_n = \alpha, \limsup s'_n = \beta$.

Metric Spaces

D7.1 Metric Space A set X whose elements are points, is a metric space if for any two points $p, q \in X$, there is an associated real number $d(p, q)$, the distance from p to q , such that it satisfies:

- Positivity: $d(p, q) > 0$ for $p \neq q, d(p, p) = 0$
- Symmetry: $d(p, q) = d(q, p)$
- Triangle Inequality: $d(p, q) \leq d(p, r) + d(r, q), r \in X$

Any function $d : X \times X \rightarrow \mathbb{R}$ satisfying these is called a distance function or metric.

E7.2 Inner Product Space A vector space V over a field $F = \mathbb{R}, \mathbb{C}$ with inner product $\langle -, - \rangle : V \times V \rightarrow F$ such that

- Conjugate Symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$. If $F = \mathbb{R}$, it is just symmetry.
- Linearity in the First Argument: $\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$.
- Positive-Definiteness: $x \neq 0, \langle x, x \rangle > 0$

We can define the norm $\| \cdot \| : V \rightarrow \mathbb{R}$ by $\|x\| = \sqrt{\langle x, x \rangle}$. Taking $X = V, d(x, y) = \|x - y\|$ defines a metric space.

E7.3 Euclidean Space Define an inner product on $\mathbb{R}^k$ for any $k \in \mathbb{N}$ by $\langle (x_1, \dots, x_k), (y_1, \dots, y_k) \rangle_{\mathbb{R}^k} = \sum_k x_k y_k$. An Euclidean space is a pair $(V, \langle -, - \rangle)$ where V is a vector space and $\langle -, - \rangle$ is a norm which is isomorphic as an inner product space to $(\mathbb{R}^k, \langle -, - \rangle_{\mathbb{R}^k})$.

D7.8 ε -neighbourhood Let (X, d_X) be a metric space. For $\varepsilon > 0, x \in X$, the ε -neighbourhood of x is $N_\varepsilon(x) := \{y \in X : d_X(x, y) < \varepsilon\}$.

D7.9 Convergence in Metric Space Let X be a metric space, and $(p_n)_{n \in \mathbb{N}}$ be a sequence in X . (p_n) converges in X to a point $p \in X$ if $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow d(p_n, p) < \varepsilon$. Alternatively, $(d(p_n, p))_{n \in \mathbb{N}} \rightarrow 0$ in $\mathbb{R}_{\geq 0}$.

L7.10 Uniqueness of Limits in Metric Spaces The limit of a sequence in a metric space, if it exists, is unique.

D7.11 A set $E \subseteq X$ is bounded if there exists $M \in \mathbb{R}, q \in X$ such that $\forall p \in E, d(p, q) < M$.

D7.12 A sequence (p_n) in X is eventually constant if $\exists N \in \mathbb{N}$ such that $\forall n \geq N$ we have $p_n = p$. It is bounded in X if $\{p_n\}$ is bounded in X .

P7.13 Let (p_n) be a sequence in a metric space. If (p_n) converges, then it is bounded.

P7.14 (p_n) converges in X iff it is eventually constant for the discrete metric.

L7.15 (p_n) converges in X iff every subsequence of (p_n) converges in X . The limit of (p_n) is equal to the limit of every one of its subsequences.

D7.16 Cauchy A sequence (p_n) in X is Cauchy if $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \in \mathbb{N}, m, n \geq N \Rightarrow d(p_n, p_m) < \varepsilon$. The sequence $(d(p_n))$ is Cauchy in $\mathbb{R}$.

P7.17 If (p_n) is convergent in X , then it is Cauchy.

P7.18 If (p_n) is Cauchy in X , it is bounded.

P7.19 If (p_n) is Cauchy in X and there exists a subsequence that converges in X , then (p_n) also converges in X to the same limit.

D7.20 A metric space X is Cauchy complete iff every Cauchy sequence in X is convergent in X .

Limits in Metric Spaces

P8.1 Convergence in Euclidean Spaces Let $V \simeq \mathbb{R}^k$ be an Euclidean space with ordered basis B of size k . For each $i \leq k$, let $\pi_i : V \rightarrow \mathbb{R}$ denote the projection onto the ith coordinate. Then any sequence (p_n) in V converges in V to $p \in V$ if for each i , the sequence $(\pi_i(p_n))$ converges in $\mathbb{R}$ to $\pi_i(p) \in \mathbb{R}$. The converse also holds.

P8.3 Any Euclidean space is Cauchy complete.

T8.4 Let V be an Euclidean space and (p_n) a bounded sequence. Then (p_n) contains a convergent subsequence.

D8.5 Let $E \subset X. p \in X$ is a limit point of E if there exists a sequence (p_n) such that $p \neq p_n \in E$ for all n and $\lim_{n \rightarrow \infty} p_n = p. p \in E$ is not required.

D8.6 Let $E \subset X, f : E \rightarrow Y$, and p be a limit point of E . We write $f(x) \rightarrow q \in Y$ as $x \rightarrow p$, or $\lim_{x \rightarrow p} f(x) = q$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $d_Y(f(x), q) < \varepsilon$ for all points $x \in E$ for which $0 < d_X(x, p) < \delta. p \in X$ but not necessarily E . We also could have $f(p) \neq \lim_{x \rightarrow p} f(x)$.

P8.7 Sequential Criterion for Limit Suppose $E \subset X, f : E \rightarrow Y, p$ is a limit point of E . Then $\lim_{x \rightarrow p} f(x) = q$ iff $\lim_{n \rightarrow \infty} f(p_n) = q$ for every sequence $\{p_n\}$ in E such that $p_n \neq p, \lim_{n \rightarrow \infty} p_n = p$.

P8.8 Suppose $E \subset X, f : E \rightarrow Y, p$ is a limit point of E . If there exists $q, q' \in Y'$ such that $\lim_{x \rightarrow p} f(x) = q, q'$, then $q = q'$.

P8.10 Let $E \subset X, f, g : E \rightarrow \mathbb{R}$, and p be a limit point of E . Suppose $\lim_{x \rightarrow p} f(x) = L$ and $\lim_{x \rightarrow p} g(x) = M, L, M \in \mathbb{R}$. Let $k \in \mathbb{R}$ be a fixed constant. Then the sum, difference, scaling, composition, and division of limits holds.

Continuity

D8.11 Continuity Suppose $(X, d_X), (Y, d_Y)$ are metric spaces, $E \subseteq X, p \in E, f: E \rightarrow Y$. Then f is continuous at $p \in E$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $d_Y(f(x), f(p)) < \varepsilon$ for all points $x \in E$ for which $d_X(x, p) < \delta$. If p is a limit point of E , then f is continuous at p iff $\lim_{x \rightarrow p} f(x) = f(p)$. If f is continuous at every point of E , then f is said to be continuous on E .

E8.12 Isometry A map $f: X \rightarrow Y$ is an isometry iff $\forall x_1, x_2 \in X, d_X(x_1, x_2) = d_Y(f(x_1), f(x_2))$. It is injective and continuous.

P8.15 Sequential Criterion for Continuity Let $(X, d_X), (Y, d_Y)$ be metric spaces and $f: X \rightarrow Y$ be a map. f is continuous at $p \in X$ iff $\lim_{n \rightarrow \infty} f(p_n) = f(p)$ for every sequence (p_n) in X such that $\lim_{n \rightarrow \infty} p_n = p$.

P8.17 Composition of Continuous Suppose X, Y, Z are metric spaces, $E \subseteq X, f: E \rightarrow Y, g: f(E) \rightarrow Z, h(x) = g(f(x))$. If f is continuous at $p \in E$ and g is continuous at $f(p)$, then h is continuous at p .

P8.18 The function $(f_1, \dots, f_k): X \rightarrow Y^k$ defined by $p \mapsto (f_1(p), \dots, f_k(p))$ is continuous iff all f_i are continuous.

P8.20 The maps $+, -, \times, (-)^{-1}$ are all continuous.

C8.21 Let $f, g: X \rightarrow \mathbb{R}$ be continuous functions on X . Then $f + g, -f, fg, \frac{1}{f}$ are all continuous.

P8.22 Let X be a metric space, V be a Euclidean space, $f, g: X \rightarrow V$ be V -valued continuous functions on $X, \alpha: X \rightarrow \mathbb{R}$ be $\mathbb{R}$ -valued continuous function on X . Then $f + g, -f, \alpha f, f \cdot g, \|f\|$ are all continuous.

D8.23 Limits to $\pm\infty$ Let $E \subseteq X, f: E \rightarrow \mathbb{R}$ be a function and p be a limit point of E . $f(x)$ tends to $+\infty$ as $x \rightarrow p$ if for every $M > 0$, there exists $\delta > 0$ such that for all $x \in X$ and $0 < d(x, p) < \delta$ we have $f(x) > M$. Then $\lim_{x \rightarrow p} f(x) = +\infty$.

P8.24 Let X be a metric space, $E \subseteq X, f: E \rightarrow \mathbb{R}, p$ be a limit point of E . Then $\lim_{x \rightarrow p} f(x) = +\infty$ iff for every sequence $\{p_n\}$ in $E \setminus \{p\}$ satisfying $\lim_{n \rightarrow \infty} p_n = p$ one has $\lim_{n \rightarrow \infty} f(p_n) = +\infty$.

Real-Valued Functions

D9.1/9.2 Let p be a limit point of $X \cap (p, \infty)$. L is the right-hand limit of f at p if for any $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in X, p < x < p + \delta \Rightarrow |f(x) - L| < \varepsilon$. Then $\lim_{x \rightarrow p^+} f(x) = L$. The left-hand limit has a similar definition.

D9.3 Let $\emptyset \subseteq X \subseteq \mathbb{R}, f: X \rightarrow \mathbb{R}$ be a function. Suppose X is not bounded above. We say that L is the limit of f as $x \rightarrow \infty$ if for any given $\varepsilon > 0$, there exists $M > 0$ such that $x \in X, x > M \Rightarrow |f(x) - L| < \varepsilon, \lim_{x \rightarrow \infty} f(x) = L$. Similarly for the not bounded below case with $x < M$.

D9.4 Let $f: X \rightarrow \mathbb{R}$ be a real-valued function. f is bounded iff the image $f(X)$ is a bounded set, i.e. there exists $M > 0$ such that $\forall x \in A, |f(x)| \leq M$. f is bounded on A if the image $f(A)$ is bounded.

P9.5 Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function on the closed bounded interval $[a, b]$. Then f is bounded on $[a, b]$.

T9.6 Extreme Value Theorem Suppose $f: [a, b] \rightarrow \mathbb{R}$ is a continuous function on the closed bounded interval $[a, b]$. Then f has an absolute maximum and absolute minimum on $[a, b]$. f must be continuous, X must be bounded, closed, and $\mathbb{R}$ must be complete.

T9.8 Intermediate Value Theorem Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous on the closed bounded interval $[a, b]$. Then for $f(a) < L < f(b)$ or the other way around, there exists $c \in (a, b)$ such that $f(c) = L$. f must be continuous, X must be connected, $\mathbb{R}$ must be complete.

P9.10 Let $a, b \in \mathbb{R}, a < b$. Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous on the closed bounded interval $[a, b]$. Then the image $f([a, b])$ is also a closed bounded interval.

L9.11 An interval is a subset I of $\mathbb{R}$ such that $\forall x, y \in I, x \leq y \Rightarrow [x, y] \subseteq I$.

P9.12 Preservation of Intervals Let I be an interval in $\mathbb{R}$, and suppose a function $f: I \rightarrow \mathbb{R}$ is continuous on I . Then $f(I)$ is an interval. It might not have the same form however.

P9.14 Continuous Inverse Theorem Let $I \subseteq \mathbb{R}$ be an interval and $f: I \rightarrow \mathbb{R}$ be a strictly monotone function. If f is continuous on I and $J = f(I)$, then its inverse function $f^{-1}: J \rightarrow \mathbb{R}$ is strictly monotone and continuous on J .

D9.16 Let X, Y be metric spaces. A function $f: X \rightarrow Y$ is uniformly continuous on X if for any given $\varepsilon > 0$, there exists $\delta > 0$ such that $x, u \in X, d_X(x, u) < \delta \Rightarrow d_Y(f(x), f(u)) < \varepsilon$. δ only depends on ε . Any uniformly continuous function is continuous.

P9.18 Sequential Criterion for Uniform Continuity Let $f: X \rightarrow Y$ be a function. TFAE: f is uniformly continuous on X ; for any two sequences $(x_n), (u_n) \in X$ such that $d_X(x_n, u_n) \rightarrow 0$, we have $d_Y(f(x_n), f(u_n)) \rightarrow 0$.

C9.19 If $X \subseteq \mathbb{R}$, then any function $f: X \rightarrow \mathbb{R}$ is not uniformly continuous on X iff there exists sequences $(x_n), (u_n) \in X$ such that $x_n - u_n \rightarrow 0$ but $f(x_n) - f(u_n) \not\rightarrow 0$.

P9.21 Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on a closed bounded interval $[a, b]$. Then f is uniformly continuous on $[a, b]$.

P9.22 Let $f: X \rightarrow Y$ be uniformly continuous. If (x_n) is Cauchy in X , then $(f(x_n))$ is Cauchy in Y .

P9.23 A function $f: (a, b) \rightarrow \mathbb{R}$ is uniformly continuous iff the limits $L_a = \lim_{x \rightarrow a} f(x)$ and $L_b = \lim_{x \rightarrow b} f(x)$ exist and $\tilde{f}: [a, b] \rightarrow \mathbb{R}$ defined by joining $f(x), L_a, L_b$ is continuous.

Topology

D10.1 Topology A topology on a set X is a collection $\mathcal{T}$ of subsets of X with the following properties:

- $\emptyset, X \in \mathcal{T}$.
- The union of elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$.
- The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

A set X for which a topology $\mathcal{T}$ has been specified is a topological space specified by $(X, \mathcal{T})$. $U \subseteq X$ is an open set of X if U belongs to collection $\mathcal{T}$.

D10.2 Closed Subsets Let $(X, \mathcal{T})$ be a topological space. $Z \subseteq X$ is closed if the complement $X - Z$ is an open set.

E10.3 The discrete topology is the power set while the trivial topology is the set and the null set.

E10.6 Finite Complement Topology
 $\mathcal{T}_f = \{U \subseteq X, X - U \text{ is finite or is all of } X\}$

D10.7 If $\mathcal{T} \subseteq \mathcal{T}'$, then $\mathcal{T}'$ is finer than $\mathcal{T}$ and $\mathcal{T}$ is coarser than $\mathcal{T}'$. The discrete topology is the finest possible topology.

E10.8 The metric topology is $\mathcal{T}_d = \{U \subseteq X : \forall p \in U, \exists r > 0, N_r(p) \subseteq U\}$

L10.9 $\forall p \in X, r > 0$, the ε -neighbourhood $N_r(p)$ is open in the metric topology.

P10.11 Let (X, d) be a metric space. $E \subseteq X$ is closed iff every limit point of E is contained in E .

P10.16 Union and Intersection of Closed Sets Let X be a topological space. Then the intersection of elements of any collection of closed sets or the union of any finite collection of closed sets is a closed set.

D10.17 Closure Let X be a topological space with $E \subseteq X$. The closure $\overline{E}$ of E is the intersection of all closed subsets of X which contain E . $E = \overline{E}$ iff E is closed.

D10.18 Interior Let X be a topological space with $E \subseteq X$. A point $p \in X$ is an interior point if there exists an open set U such that $p \in U, U \subseteq E$. The interior of E , $\text{Int}E$, is the union of all interior points of E .

X10.19 A subset E is open iff every point $x \in E$ is an interior point.

Basis

P11.1 Intersection of Topologies For any non-empty collection $\mathcal{Z} = \{\mathcal{T}_i\}_{i \in I}$ of topologies on X , the intersection $\mathcal{T} = \bigcap_{i \in I} \mathcal{T}_i$ is also a topology on X .

D11.2 Let $S \subseteq \mathcal{P}(X)$ be any collection of subsets of X . Consider the collection $\mathcal{J}_S$ of all topologies $\mathcal{T}$ on X with S which is nonempty. Then $\mathcal{T}_S = \bigcap_{\mathcal{T} \in \mathcal{J}_S} \mathcal{T}$ is the topology generated by S , and is the coarsest topology on X where every member of S is open.

D11.5 Basis Let $\mathcal{B}$ be a collection of subsets of X such that:

- For each $x \in X$, there is at least one $B \in \mathcal{B}$ such that $x \in B \Leftrightarrow X = \bigcup_{B \in \mathcal{B}} B$.
- If $B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2$, then there is $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Then we claim the topology $\mathcal{T}_{\mathcal{B}}$ generated by $\mathcal{B}$ is as follows: A subset U of X is open iff for each $x \in U$, there is $B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U$. $\mathcal{B}$ is a basis for the topology $\mathcal{T}_{\mathcal{B}}$.

C11.6 Let X be a set and $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on X . Then $\mathcal{T}$ is the collection of all unions of elements of $\mathcal{B}$.

E11.7 A basis for the metric topology of X is $\mathcal{B} = \{N_r(p) \in \mathcal{P}(X) : p \in X, r \in \mathbb{R}_{>0}\}$, all open neighbourhoods in X .

Continuity and Homeomorphisms

D11.8 Topological Continuity Let X, Y be topological spaces. $f: X \rightarrow Y$ is continuous if for each open subset V of $Y, f^{-1}(V) = \{x \in X : f(x) \in V\}$ is an open subset of X .

P11.10 If $f: X \rightarrow Y, g: Y \rightarrow Z$ are continuous, then $g \circ f: X \rightarrow Z$ also is.

P11.12 Suppose X and Y are metric spaces with metrics d_X, d_Y . $f: X \rightarrow Y$ is continuous wrt the metric topologies iff $\forall p \in X, \forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x \in X, d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) < \varepsilon$.

D11.13 Subspace Topology Let X be a topological space with topology $\mathcal{T}$. If A is a subset of $X, \mathcal{T}_A = \{A \cap U : U \in \mathcal{T}\}$ is the subspace topology on A .

P11.14 If A is a subspace of X , the inclusion function $j: A \rightarrow X$ is continuous. The subspace topology is the coarsest topology on X such that the inclusion map is continuous.

P11.16 If $f: X \rightarrow Y$ is continuous and if A is a subspace of X , then the restricted function $f|_A: A \rightarrow Y$ is continuous using $f \circ j$.

P11.17 Let $f: X \rightarrow Y$ be continuous. If Z is a topological space having Y as a subspace, then $h: X \rightarrow Z$ obtained by expanding the range of f is continuous using $h = i \circ f$.

P11.18 Let X, Y be topological spaces, and $A \subseteq Y$ be a subset with subspace topology. For any map $f_0: X \rightarrow A$, TFAE: The map is continuous, or the composition $i \circ f_0: X \rightarrow Y$ is continuous where $i: A \rightarrow Y$ is the inclusion map.

D11.19 Homeomorphism A continuous map $f: X \rightarrow Y$ is a homeomorphism iff there exists a continuous map $g: Y \rightarrow X$ such that $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. It must be bijective.

D11.21 Homeomorphic Topological Subspaces X, Y are homeomorphic iff there exists a homeomorphism $f: X \rightarrow Y$. Properties of topological spaces invariant under homeomorphisms are called topological properties.

Connectedness

D12.1 A separation of X is a pair U, V of disjoint nonempty open subsets of X whose union is X . X is said to be connected iff there does not exist a separation of X .

D12.3 $A, B \subseteq X$ are separated if both $A \cap \bar{B}, \bar{A} \cap B$ are empty.

P12.4 $E \subseteq X$ is connected if E is not a union of two nonempty separated sets.

E12.5 If the topology of X is trivial, then it is connected. If X is empty or a singleton, it is connected. If X is discrete, it is connected iff it is empty or a singleton.

E12.6 The connected subsets of $\mathbb{R}$ are exactly the intervals, so $\mathbb{R}$ is connected.

R12.7 A subspace of a connected space need not be connected.

T12.8 Intermediate Value Theorem Let X, Y be topological spaces. If X is connected and $f: X \rightarrow Y$ is a continuous map, then $f(X)$ is connected.

C12.10 If X, Y are homeomorphic, X is connected iff Y is connected.

D12.12 Path Connected A path in X from x to y is a continuous map $f: [a, b] \rightarrow X$ of some closed interval in the real line into X such that $f(a) = x, f(b) = y$. A space X is path connected if every pair of points of X can be joined by a path in X .

P12.13 Every path connected space X is connected. The converse is not true.

D12.16 For $x, y \in \mathbb{R}^k$, define $[x, y] := \{\lambda x + (1 - \lambda)y : \lambda \in [0, 1]\}$. It is connected. $E \subseteq \mathbb{R}^k$ is convex if $\forall x, y \in E, [x, y] \subseteq E$.

P12.18 Convex subsets of $\mathbb{R}^k$ are connected.

Hausdorff Spaces, Compactness

D12.19 Open Neighbourhood Let $x \in X$. An open subset $U \subseteq X$ which contains x is an open neighbourhood of x .

D12.20 Convergence in Topological Spaces Let (x_n) be a sequence in X and $x \in X$. (x_n) converges to $x \in X$ if for every open neighbourhood U of x , there exists $N \in \mathbb{N}$ such that for all $n \geq N, x_n \in U$.

P12.21 If X is a Hausdorff space, then any sequence (x_n) in X converges to at most one point.

D12.24 Hausdorff X is a Hausdorff space if for each pair $(x_1, x_2) \in X, x_1 \neq x_2$, there exists open neighbourhoods U_1, U_2 of x_1, x_2 respectively such that $U_1 \cap U_2 = \emptyset$.

E12.26 If X is a metric space, then X is Hausdorff.

P12.27 A subspace of a Hausdorff space is a Hausdorff space.

D12.28 The property of being Hausdorff is a topological property.

D12.29 A collection $\mathcal{A}$ of subsets of X covers X iff the union of the elements of $\mathcal{A}$ is equal to X . It is an open covering if $U \in \mathcal{A}$ is open.

D12.30 Compactness X is compact iff every open covering $\mathcal{A}$ of X contains a finite subcollection that also covers X .

E12.31 If the topology of X is trivial then X is compact.

E12.32 If the topology of X is discrete then X is compact iff X is finite.

E12.33 Any finite topological space X is compact.

E12.34 $\mathbb{R}$ with the standard topology is not compact.

E12.35 $[a, b]$ is compact.

E12.36 $X = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ is compact.

D12.37 $Y \subseteq X$ is compact if Y with the subspace topology is compact. A collection $\mathcal{A}$ of subsets of X covers Y if $\bigcup_{U \in \mathcal{A}} U \supseteq Y$, and Y is compact if every open cover by sets in X has a finite subcover.

R12.38 The subspace of a compact space need not be compact.

P12.39 A finite union of compact subsets is compact.

P12.40 Every closed subspace of a compact space is compact.

P12.41 Every compact subspace of a Hausdorff space is closed.

L12.42 If Y is a compact subspace of a Hausdorff space X and $x_0 \notin Y$, then there exists disjoint open sets U, V with $x_0 \in U, Y \subseteq V$.

E12.43 For disjoint compact sets A, B in Hausdorff X , there exists disjoint open $U \supseteq A$ and $V \supseteq B$.

E12.44 In $\mathbb{R}$, only the closed interval $[a, b]$ is compact.

E12.46 In finite-complement topology, all subsets are compact but only finite sets are closed. This is not Hausdorff if X is not finite.

D12.47 Sequential Compactness A topological space X is said to be sequentially compact if every sequence in X has a convergent subsequence whose limit is X . $[a, b]$ satisfies this.

E12.48 The closed bounded interval $[a, b]$ is sequentially compact.

T12.49 Let X be a metric space. Suppose $A \subseteq X$ is sequentially compact. Then A is closed and bounded in X .

Compact subsets of $\mathbb{R}^n$

T13.1 Heine-Borel Theorem A subspace A of $\mathbb{R}^n$ is compact iff it is closed and bounded.

P13.2 If X is a metric space, then X is compact iff it is sequentially compact.

E13.3 The unit sphere S^{n-1} and closed unit ball B^n in $\mathbb{R}^n$ are compact because they are closed and bounded.

T13.4 Extreme Value Theorem Let X be any compact topological space and $f: X \rightarrow Y$ be continuous. Then the image $f(X)$ is compact.

R13.5 When $Y = \mathbb{R}, X = [a, b]$, since X is compact, $f(X)$ is compact and hence closed and bounded. So $\alpha = \sup f(X)$ exists, and since $f(X)$ is closed, $\alpha \in f(X)$, so there exists some $c \in [a, b]$ such that $f(c) = \alpha$. This gives the maximum.

C13.6 Compactness is a topological property.

Random Tutorial Crap

Bernoulli's Inequality $\forall x \geq -1, \forall n \in \mathbb{N}, (1+x)^n \geq 1+nx$

T1P7 $\sup X = -\inf -X$

T2P4 $\lim_{n \rightarrow \infty} |x_n| = 0$ iff $\lim_{n \rightarrow \infty} x_n = 0$.

T3P5 x_n converges to $x \in \mathbb{R}$ iff x_{n_k} converges to x for every subsequence of x_n .

T3P6 If every subsequence of x_n has a subsequence which converges to 0, then x_n converges and $\lim_{n \rightarrow \infty} x_n = 0$.

T4P1 $\lim_{n \rightarrow \infty} \cos n$ does not exist.

T4P6 $\limsup (x_n + y_n) \leq \limsup x_n + \limsup y_n$, but equality might not hold.

T7P3 $\varepsilon - \delta$ Let $\varepsilon > 0$ be given.

$$\begin{aligned} \left| \frac{(2x+1)(x-2)}{(6x+2)} - (-1) \right| &= \left| \frac{2x^2 - 3x - 2 + 6x + 2}{6x + 2} \right| \\ &= \frac{|x||2x+3|}{2|3x+1|} \end{aligned}$$

Restrict x to $0 < |x-0| < \frac{1}{4}$. Then $\frac{5}{2} < 2x+3 < \frac{7}{2}, \frac{1}{4} < 3x+1 < \frac{7}{4}$.

$|2x+3| < \frac{7}{2}, |3x+1| > \frac{1}{4}$. Then

$\left| \frac{(2x+1)(x-2)}{(6x+2)} - (-1) \right| < \frac{7/2}{2 \cdot 1/4} |x| = 7|x-0|$. Now set $\delta = \min(\frac{1}{4}, \frac{\varepsilon}{7}) > 0$.

Then $\left| \frac{(2x+1)(x-2)}{(6x+2)} - (-1) \right| < 7 \times \frac{\varepsilon}{7} = \varepsilon$. For numerator, pick upper bound, for denominator, pick lower bound.

T7P5 Let $f: A \rightarrow \mathbb{R}$ be a function and a be a limit point of A . If $\lim_{x \rightarrow a} |f(x)| = 0$ then $\lim_{x \rightarrow a} f(x) = 0$

T8P3 $h(x) = (x - [x]) \sin(\pi x)$ is continuous although $(x - [x])$ is not.

T8P6 Suppose $h(x) = \frac{1}{q}, x \in \mathbb{Q} \cap (0, +\infty)$ and x is rational. We can use it to obtain (x_n) of irrational positive numbers that converge to $a \in \mathbb{Q} \cap (0, +\infty)$.

T9P5 Show there exists a point where $f(c) = f(c+3) = f(6)$. Let $g(x) = f(x) - f(x+3)$. Then $g(0) = f(0) - f(3), g(3) = f(3) - f(6) = f(3) - f(0) = -g(0)$. 0 must be a value between $g(0), g(3)$. Then by IVT, there exists c such that $g(c) = f(c) - f(c+3) = 0$.

T9P7 Suppose g is continuous and injective on $[0, 5]$, and $g(0) > g(5)$. Show g is strictly decreasing on $[0, 5]$. Claim $g(0) > g(x) > g(5), 0 < x < 5$. Since g

is injective, $g(x) \neq g(0), g(5)$. Suppose that does not hold, then either $g(x) > g(0)$ or $g(x) < g(5)$. If $g(x) > g(0) > g(5)$, by IVT, there exists $x' \in (x, 5), g(x') = g(0)$, contradiction. If $g(0) > g(5) > g(x)$, by IVT, there exists $x' \in (0, x), g(x') = g(5)$, contradiction. To prove g is strictly decreasing, take $x, u \in [0, 5], x < u$. Suppose $g(x) < g(u)$. Since $g(0) > g(u) > g(x)$, by IVT, there exists $x' \in (0, x), g(x') = g(u)$, contradiction. So $g(x) \geq g(u)$, and since $g(x) \neq g(u)$, g must be strictly decreasing.

T10P4 The Lipschitz condition says that

$\exists K > 0, |f(x) - f(y)| \leq K|x - y|, \forall x, y \in A$. Choose $\delta = \frac{\varepsilon}{K} > 0$. Then $x, u \in A, |x - u| < \delta \Rightarrow |f(x) - f(y)| \leq K|x - y| < K \cdot \delta = K \cdot \frac{\varepsilon}{K} = \varepsilon$, and f is uniformly continuous on A . Find the choice of δ based on ε and distance.

T11P1 $\mathbb{R} - \mathbb{N} = \bigcup_{z \in \mathbb{N}} (z-1, z) \cup (-\infty, 0)$ so $\mathbb{R} - \mathbb{N}$ is open and $\mathbb{N}$ is closed.

T11P2 Let $f: X \rightarrow \mathbb{R}$ be a continuous function and $A \subseteq X$ be open. If $f(A)$ is a single point, then $f(A)$ is also a single point.

T11P3 If A, B are disjoint compact subspaces of the Hausdorff space X , then there exists disjoint open sets U, V containing A, B respectively.

T11P5 Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of connected subspaces of X , such that $A_n \cap A_{n+1} \neq \emptyset$. Then $\bigcup A_n$ is connected.

T11P6 No pair of $(0, 1), (0, 1], [0, 1]$ are homeomorphic. Only $[0, 1]$ is compact. $(0, 1) = \{g(1)\}$ is not connected.

T11P7 Let $f: X \rightarrow \mathbb{R}$ be a continuous map where X is the unit circle outline. Then there exists a point $(x, y) \in X, f(x, y) = f(-x, -y)$.

Extra P1 $\mathbb{Q}$ is neither open or closed in $\mathbb{R}$. $(q - \varepsilon, q + \varepsilon)$ always contains irrationals so no point $q \in \mathbb{Q}$ has a small open interval around it that sits entirely in $\mathbb{Q}$, so $\mathbb{Q}$ is not open. A set is closed if it contains all of its limit points. There exists a sequence of rationals that tends to $\sqrt{2} \notin \mathbb{Q}$, so $\mathbb{Q}$ is not closed.

Extra P2 $\mathbb{Q}$ satisfies $\text{Int}(\mathbb{Q}) = \emptyset, \bar{\mathbb{Q}} = \mathbb{R}$.

Extra P3 If $F \subseteq \mathbb{R}$ is closed, nonempty and bounded above, then $\sup F \in F$.

Extra P6 A finite union of compact subspaces of X is compact.

Extra P8 The intersection of an arbitrary collection of compact sets in $\mathbb{R}^k$ is compact.

Extra P10 If $f: X \rightarrow Y$ is continuous, where X is compact Y is Hausdorff, then f is a closed map (from a closed set to another closed set).

Review 1 If $f: X \rightarrow Y$ is continuous, then for every open $U \subseteq X, f(U)$ is open in Y is not necessarily true.

Review 2 $f(x) = \sqrt{x-1}$ is uniformly continuous on $[1, \infty)$. We have $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x-y|}, |f(x) - f(y)| = |\sqrt{x-1} - \sqrt{y-1}| \leq \sqrt{|(x-1) - (y-1)|} = \sqrt{|x-y|} < \sqrt{\delta} = \varepsilon$ for $\delta = \varepsilon^2$.

It can be shown it without picking any particular δ , it's continuous.